

1. Visual Work Has a Different Evidence Surface

A page can pass tests and still be visibly broken. Layout, hierarchy, clipping, animation timing, responsive behavior, and interaction feedback are properties of a rendered state. For these tasks, code inspection is necessary but not sufficient.

Astra's computer and browser capabilities make visual verification part of the same engineering loop rather than a separate manual afterthought.

2. The Render-Inspect-Change Loop

1. Render the current state at the relevant viewport or application state.
2. Inspect the artifact itself, not only the source.
3. Attach feedback to a visible element or exact state when possible.
4. Change the smallest cause that explains the defect.
5. Re-render the same state.
6. Compare and then inspect adjacent states likely to regress.

The loop is analogous to a failing test. A screenshot before the change establishes the visual failure; the same view after the change supplies direct evidence.

3. The Page as Specification

Jason Liu describes using Codex's in-app browser to create an `index.html`, open it, and annotate elements directly. The annotation carries screenshot and element context into the thread. This makes the page itself a shared specification.

The important principle is locality of critique. "This control needs more room" attached to the control is more informative than a paragraph trying to identify it indirectly. Context becomes grounded in the artifact.

4. Reference Fidelity

Kundel reports that Astra can infer substantial product behavior from rough ideas and strong references. This creates both leverage and risk. The more the reference determines, the less the prompt should invent. But the agent still needs to know which properties are normative.

A visual reference can constrain typography, spacing, hierarchy, density, interaction model, or motion. Say which relation matters when ambiguity would change the result. Do not ask for "the vibe" when the actual requirement is a 12-column rhythm or a specific keyboard interaction.

5. Anti-Slop Criterion

Generic agent-made interfaces often reveal themselves through unnecessary cards, gradients, oversized headings, decorative copy, repeated explanatory labels, and controls added because a dashboard convention exists rather than because the task needs them. The corrective principle is functional evidence: every visible element should justify itself by a user action, information need, or source reference.

If removing an element does not impair the task or the specified reference, remove it. Visual restraint is not minimalism as fashion; it is refusal to invent unsupported interface decisions.

6. Native Applications

For 3D or media work, the best verification surface may be the domain application itself. Kundel's Astra experiments used Blender even when the final output belonged in a browser. A native tool can expose geometry, materials, timeline state, or render settings more directly than a generic web preview.

The rule is still narrowest sufficient tool. Use the specialized application because it represents the object better, not because cross-app work sounds impressive.

7. Visual Acceptance Criteria

Good visual criteria name states:

- target viewport sizes;
- required content and hierarchy;
- reference screenshots or design files;
- keyboard and pointer interactions;
- empty/loading/error states when relevant;
- prohibited overflow, clipping, or unintended scroll;
- motion or reduced-motion expectations.

Then verify those states. "Looks good" is not an acceptance test.

References: Jason Liu, "Three Ways Codex Can Use a Computer"; Dominik Kundel, "Five things I learned from using Astra"; OpenAI, Codex Browser and Computer Use documentation.

8. Worked Case: Responsive Header

Suppose a header clips at 390 pixels. The source shows a fixed-width search field beside two actions. The first screenshot establishes the failure. Inspect computed layout and discover that the search field has a 320-pixel minimum width inherited from desktop. Change the responsive constraint rather than hiding an action. Re-render at 390 pixels, then 375 and 768 to test the boundary around the change. Keyboard through the controls to ensure the new wrapping order still matches reading order.

9. Visual Diff Limits

Pixel comparison can reveal unintended change but cannot judge every improvement. Font rendering, antialiasing, timestamps, and dynamic data can create harmless diffs. Conversely, two screenshots can look similar while keyboard behavior is broken. Use image comparison as one evidence source and pair it with semantic interaction checks when the task includes behavior.

10. Reference Hierarchy

When several references conflict, rank them by authority for the property in question. A current design file may control layout; production copy may control text; the existing implementation may control behavior not yet represented in design. The agent should not average contradictions into a new hybrid. Surface the conflict or follow the source that actually owns the decision.

Independent Open Course

<https://github.com/dicnunz/codex-astra-course>

CODEX.00 | Codex with GPT-6 Astra

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